

DESIGN THINKING FOR SOFTWARE ENGINEERING

(LAB – 2 REPORT)

TOPIC : E-LEARNING PLATFORMS FAIL TO KEEP LEARNERS ENGAGED

STEP 1: Define the Target User

- **User Group:** Online learners / students
 - **Age:** 18–25 years
 - **Education:** Undergraduate / Diploma students
 - **Device Used:** Smartphone / Laptop
 - **Digital Literacy:** Medium to High
 - **Platforms Used:** Coursera, Udemy, Google Classroom, YouTube Learning, LMS
 - **Learning Mode:** Self-paced online learning
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STEP 2: List Initial Assumptions

Assumption Table

- **A1:** Learners lose interest quickly in online courses
 - **A2:** Long video lectures reduce attention
 - **A3:** Learners feel isolated in e-learning
 - **A4:** Lack of interaction causes disengagement
 - **A5:** Courses are too theoretical and boring
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STEP 3: Prepare Interview Questions

- What e-learning platforms do you use regularly?
 - What makes an online course boring for you?
 - How long can you stay focused during online classes?
 - What problems do you face while learning online?
 - Do you complete most online courses? Why or why not?
 - What features would make learning more interesting?
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STEP 4: Conduct User Interview

Observations:

- Learners lose focus after 15–20 minutes



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- Many skip videos or play at higher speed
- Limited interaction with instructors
- Learners multitask during sessions

Sample User Statements:

- "Videos are too long and boring."
- "There is no motivation to complete courses."
- "I feel disconnected from teachers."

STEP 5: Create Empathy Map

1. Says

- "Online classes feel monotonous."
- "I miss real classroom interaction."

2. Thinks

- Learning online is less effective
- Completion is not compulsory

3. Does

- Skips lectures
- Watches videos passively
- Drops courses midway

4. Feel

- Bored
- Unmotivated
- Distracted

STEP 6: Validate Assumptions

Assumption	Validated?	Reason
Learners lose interest	✓	Reported boredom
Long videos reduce focus	✓	Attention drops fast
Learners feel isolated	✓	Miss interaction
Learners dislike theory	✓	Prefer practical content
Learners hate technology	✗	Comfortable with tools



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STEP 7: Identify User Insights

- Engagement matters more than content quantity
- Short, interactive sessions improve focus
- Lack of feedback reduces motivation
- Social learning increases participation

✗ *"Students are lazy"*

✓ *"Students disengage when learning feels passive"*

STEP 8: Create User Persona

Name: Rahul

Age: 21

Education: Engineering Student

Device: Laptop & Smartphone

Goals

- Learn skills efficiently
- Score well in exams
- Get job-ready

Pain Points

- Long boring lectures
- No doubt-clearing
- No peer interaction

Motivation

- Career growth
- Skill development

Quote:

"I want engaging content, not just recorded videos."

STEP 9: Translate Insights to Software Needs

Insight

Software
Requirement

Low attention span Short videos (5–10)



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Insight	Software Requirement
	mins)
Need interaction	Live chats, polls, quizzes
Lack of motivation	Gamification, badges
Isolation	Discussion forums
Passive learning	Practical tasks & projects

STEP 10: Reflection & Documentation

- **What surprised you?**
 - Learners prefer interaction over content length
 - **Which assumption was wrong?**
 - Learners are not lazy; content design is weak
 - **How did empathy change thinking?**
 - Focus shifted from teaching to learner experience
 - **Effect on software design:**
 - More interactive, engaging, and learner-centric platforms
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